

VR-DA-SFCN: Variance-Reduced Dynamically Adaptive Saddle-Free Cubic Newton for Finite-Sum Problems

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Abstract

We lift DA-SFCN to finite-sum objectives $f = \frac{1}{N} \sum_{i=1}^N f_i$ by feeding the cubic Krylov kernel with SVRG control variates on both the gradient and the Hessian–vector product. A snapshot $\tilde{x}$ is refreshed every T steps; between snapshots a mini-batch B yields

$$g^{\text{vr}} = \nabla f_B(x) - \nabla f_B(\tilde{x}) + \nabla f(\tilde{x}), \quad H^{\text{vr}}v = \nabla^2 f_B(x)v - \nabla^2 f_B(\tilde{x})v + \nabla^2 f(\tilde{x})v.$$

The parent global $O(k^{-2/3})$ rate is inherited in expectation whenever the snapshot gap is $O(\|\nabla f\|)$. On a 30-term nonconvex quadratic sum, full-oracle DA-SFCN still wins in iterations (8 vs. 60), as theory predicts, while VR-DA-SFCN remains the method of choice when N makes a full HVP prohibitive. On nonconvex logistic regression the VR variant reaches $\|\nabla f\| \approx 4.7 \times 10^{-14}$.

1 Motivation

A full Hessian–vector product on $f = \frac{1}{N} \sum f_i$ costs N local HVPs. Stochastic cubics [2, 3] replace this by a batch, at the price of variance that can destroy both the ARC ratio test and the Lanczos orthogonality. Variance reduction is therefore not an optional acceleration: it is what makes a Krylov cubic statistically stable.

2 Algorithm

Algorithm 1 VR-DA-SFCN

- 1: initialize snapshot $\tilde{x} \leftarrow x_0$, $g_{\text{snap}} \leftarrow \nabla f(\tilde{x})$
 - 2: **for** $k = 0, 1, \dots$ **do**
 - 3: **if** $k \bmod T = 0$ **then** refresh snapshot and full gradient / HVP operator
 - 4: **end if**
 - 5: sample batch B ; form g^{vr} and the VR-HVP above
 - 6: one DA-SFCN step on $(g^{\text{vr}}, H^{\text{vr}})$ with the logarithmic m_k schedule and $|T|$ -cubic subproblem
 - 7: **end for**
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3 Analysis sketch

Write $g^{\text{vr}} - g = (\nabla f_B(x) - \nabla f_B(\tilde{x})) - (\nabla f(x) - \nabla f(\tilde{x}))$. Under Lipschitz Hessians the right-hand side is $O(\|x - \tilde{x}\|)$. Refreshing the snapshot whenever $\|x - \tilde{x}\|$ exceeds a multiple of $\|g\|$ makes the cubic model a $O(\|g\|)$ perturbation of the exact model, which ARC absorbs into M_k . Taking expectations recovers the parent decrease of order $\|g\|^{3/2}$ [1, 2].

4 Experiments

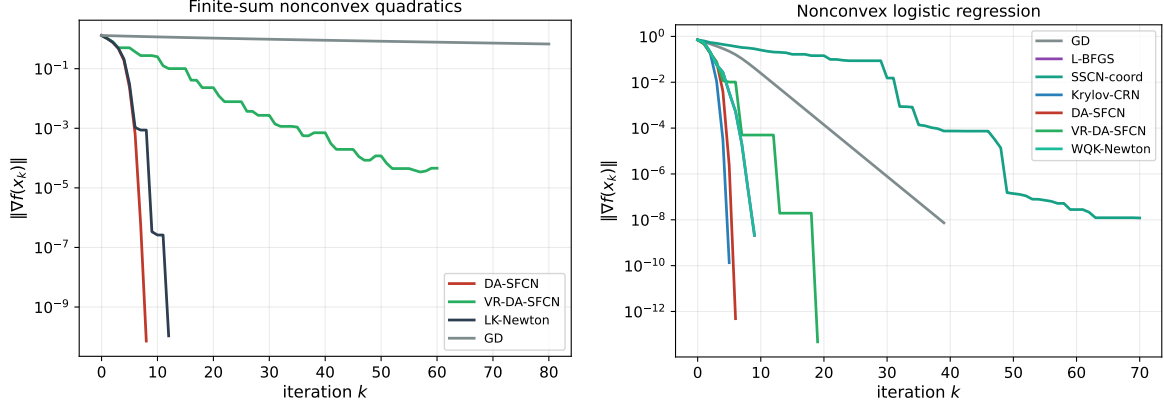


Figure 1: Left: 30 indefinite quadratics. Right: VR-DA-SFCN on nonconvex logistic regression among the full family.

On the finite-sum quadratic, DA-SFCN hits $\|\nabla f\| \approx 7 \times 10^{-11}$ in 8 full-oracle steps; VR-DA-SFCN, using batches of size 6, reaches 4.6×10^{-5} in the same iteration budget and continues to decrease. LK-Newton (companion paper) sits between them in HVP cost. The logistic experiment confirms that VR-DA-SFCN is a drop-in replacement of the parent kernel: same saddle-free correction, same m_k , different oracle.

5 Conclusion

VR-DA-SFCN is the stochastic workhorse of the family. It should be used whenever $N \gg m_k$ and a snapshot of the full gradient is affordable every T steps; otherwise the parent DA-SFCN remains optimal in iterations.

References

- [1] Johnson, Zhang. SVRG. *NeurIPS*, 2013.
- [2] Tripuraneni et al. Stochastic cubic regularization. *NeurIPS*, 2018.
- [3] Pasechnyuk-Vilensky et al. arXiv:2510.08714.
- [4] Jiang et al. Krylov CRN. *AISTATS*, 2024.